

MOSS-DOC-002
REVISION A

MOSS KIT

PAYLOAD HARDWARE

COMPONENT DESCRIPTION AND ASSEMBLY MANUAL

ATLAS-1 High-Altitude Balloon Program
University of Nevada, Las Vegas
PROVE Initiative
Prepared by the UNLV ATLAS-1 Student Team
July 2026

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RECORD OF REVISIONS

Rev.	Date	Description of Change	Approved By
—	2026-06	Initial working draft (prose format).	UNLV ATLAS-1
A	2026-07	Initial formal release. Restructured into numbered sections per MOSS-DOC-001 conventions. Scope expanded to the full ATLAS-1 sensor suite.	Noemi Magallanes

APPROVAL

Role	Name	Signature / Date
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NOTICE This document is issued as Revision A and is subject to revision. The controlled electronic copy is authoritative. Verify revision status before use.

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1.0 SCOPE

1.1 Identification

- 1.1.1** This document is the Payload Hardware Component Description and Assembly Manual for the MOSS Kit (Modular Outreach Starter Satellite), a reproducible high-altitude balloon (HAB) telemetry payload developed under the ATLAS-1 program of the PROVE initiative at the University of Nevada, Las Vegas.
- 1.1.2** The MOSS Kit is a student training platform constructed from commercial off-the-shelf components. It is intended to be reproduced by student teams in support of progressively more advanced space research activities.

1.2 Purpose

- 1.2.1** This document identifies each hardware component of the MOSS Kit flight segment, defines its function and interface, and prescribes the mechanical assembly sequence by which the components are integrated into a flight-ready payload stack.
- 1.2.2** This document is intended for student personnel assembling a MOSS Kit. No prior experience with the Raspberry Pi platform, electronics assembly, or radio telemetry is assumed.
- 1.2.3** This document addresses mechanical assembly only. Operating system preparation, interface enablement, driver installation, software configuration, and operating procedures are prescribed by MOSS-DOC-001 and are outside the scope of this document.

1.3 Design Rationale

- 1.3.1** A flight-rated CubeSat costs tens of thousands of dollars and must survive the orbital environment, placing it beyond the reach of most classrooms. The MOSS Kit reduces the concept to its essentials at a target cost of approximately \$500, such that a student may build, break, repair, and understand a working satellite payload without a launch vehicle or an institutional budget.
- 1.3.2** The MOSS Kit is not flight rated and shall not be used for an orbital launch. The kit is qualified for high-altitude balloon flight only.
- 1.3.3** The kit is modular by design. Student teams may extend the sensor suite or substitute components to suit a specific payload objective, subject to the interface constraints of Section 4.0.

1.4 Document Conventions

- 1.4.1** The verb "shall" denotes a requirement. The verb "will" denotes a statement of fact or a declaration of intent. The verb "should" denotes a recommendation. The verb "may" denotes permission.
- 1.4.2** Values enclosed in angle brackets, such as <username>, are placeholders. The operator shall substitute values selected by the operating team.

1.4.3 Component addresses stated in this document are the default values for the reference breakout boards. They are expected values, not required values. The bus scan of MOSS-DOC-001 Section 7.2 reports the actual configuration of the operator's hardware.

1.5 Applicable Documents

1.5.1 The following documents form a part of this manual to the extent specified herein. In the event of conflict, this document takes precedence for MOSS Kit assembly procedures; the referenced documents take precedence for their respective products.

Table 1-1. Applicable Documents

Reference	Title / Source
MOSS-DOC-001	MOSS Kit Payload and Ground Station System Description and Operations Manual
RPi-HW	Raspberry Pi 3 Model B+ product documentation, raspberrypi.com
WS-SENSE	Waveshare Sense HAT (B) product documentation
WS-UPS	Waveshare UPS HAT E product documentation
ADA-CP	Adafruit CircuitPython sensor driver documentation
RPi-CAM	Raspberry Pi Camera Module v2 product documentation

2.0 ACRONYMS AND DEFINITIONS

2.1 Acronyms

Acronym	Definition
CSI	Camera Serial Interface
ESD	Electrostatic Discharge
GPIO	General-Purpose Input/Output
HAB	High-Altitude Balloon
HAT	Hardware Attached on Top (Raspberry Pi expansion board)
I2C	Inter-Integrated Circuit (two-wire serial bus)
IMU	Inertial Measurement Unit
LoRa	Long Range (radio modulation scheme)
SoC	State of Charge
UART	Universal Asynchronous Receiver/Transmitter (serial port)
UPS	Uninterruptible Power Supply

2.2 Definitions

- 2.2.1 Stack.** The mechanically joined assembly of the UPS HAT, the Raspberry Pi, and the Sense HAT, mated through the 40-pin GPIO header.
- 2.2.2 HAT.** An expansion board conforming to the Raspberry Pi Hardware Attached on Top specification, which mates directly to the 40-pin GPIO header rather than connecting by cable.
- 2.2.3 Flight rated.** Tested and certified for the temperature extremes, pressure changes, and vibration environment of the intended flight regime.
- 2.2.4 Payload.** The complete assembled flight segment: the stack, the camera, the radio, and the battery system that ascend with the balloon.
- 2.2.5 Standoff.** A threaded mechanical spacer that maintains a fixed separation between two boards and carries the mechanical load of the stack.

3.0 HARDWARE SYSTEM DESCRIPTION

3.1 Architecture

- 3.1.1** The MOSS Kit flight segment is organized around a single flight computer. All sensing, imaging, power management, and radio functions are subordinate to that computer, which executes the flight software, coordinates every peripheral, and writes the authoritative data set to local storage.
- 3.1.2** Peripherals attach by one of three interfaces. The sensor suite and the battery gauge share a single I2C bus, distinguished from one another by device address. The radio occupies a dedicated UART. The camera occupies a dedicated CSI port. The rationale for this division is stated in 3.3.
- 3.1.3** The MOSS Kit is supplied in two hardware configurations. The baseline configuration employs the Waveshare Sense HAT (B), which consolidates the sensor suite onto a single stacking board and requires no discrete wiring. The ATLAS-1 reference configuration employs discrete breakout sensors on a common I2C bus, permitting individual sensor substitution at the cost of additional wiring. Both configurations present the same interfaces to the flight software.

Table 3-1. MOSS Kit Flight Segment Hardware

Item	Function	Flight Rated
Raspberry Pi 3 Model B+	Flight computer	No
Waveshare UPS HAT E	Power management; battery voltage, current, and state-of-charge monitoring	No
Waveshare Sense HAT (B)	Consolidated environmental and motion sensor suite (baseline configuration)	No
BME280	Temperature, barometric pressure, humidity, derived altitude	No
LTR390	UV and ambient light	No
TSL2591	Lux, infrared, and visible light	No
ICM20948	Acceleration, angular rate, magnetic field (IMU)	No
SX126x LoRa transceiver	One-way telemetry downlink	No
Raspberry Pi Camera Module v2	Imaging	Yes

NOTE The Camera Module v2 is the only flight-rated component in the kit. All remaining components are commercial off-the-shelf parts qualified by the program for HAB flight only. The kit as a whole shall not be represented as flight rated.

3.2 Component Functions

- 3.2.1** Raspberry Pi 3 Model B+. The flight computer. It executes the flight software, coordinates all peripheral boards, logs every record to non-volatile storage, and determines what occurs and when. Every other

component in the payload either mates to this board or is managed by it. It presents a 40-pin GPIO header, a dedicated CSI camera port, and a UART.

- 3.2.2** **Waveshare UPS HAT E.** The power management board. It houses and manages the Li-ion cells, supplies regulated power to the stack, and protects against voltage dips and brief interruptions. It additionally reports battery voltage, current, and state of charge to the flight computer over I2C at address 0x2D, such that battery condition appears in every telemetry record.
- 3.2.3** **Waveshare Sense HAT (B).** The consolidated sensor board of the baseline configuration. It mates to the GPIO header and provides environmental and motion data: temperature, humidity, barometric pressure, and an IMU comprising accelerometer, gyroscope, and magnetometer. It performs no processing of its own; all data is read by the flight computer over I2C.
- 3.2.4** **Discrete sensor suite.** The ATLAS-1 reference configuration substitutes four discrete breakout sensors for the Sense HAT, wired to the same I2C bus. The BME280 provides temperature, pressure, and humidity, from which altitude above the launch site is derived. The LTR390 provides UV and ambient light. The TSL2591 provides lux, infrared, and visible light. The ICM20948 provides acceleration and angular rate.
- 3.2.5** **SX126x LoRa transceiver.** The telemetry downlink. It transmits a 50-byte binary packet over a one-way radio link at a nominal 2-second interval. It requires exclusive use of the UART. No uplink capability exists and no acknowledgment is returned to the payload.
- 3.2.6** **Raspberry Pi Camera Module v2.** The imaging payload. It captures still photographs during flight, including horizon and downward-facing frames. It connects by ribbon cable to the dedicated CSI port and does not stack.

3.3 Interface Allocation Rationale

- 3.3.1** I2C is a shared two-wire bus. Every sensor connects to the same two pins and is distinguished solely by its device address. This permits the entire sensor suite to be served by one pair of pins, at data rates appropriate to environmental sampling.
- 3.3.2** The radio is allocated a dedicated UART because it requires exclusive use of the serial hardware. The consequence of failing to release that hardware from the operating system is stated in MOSS-DOC-001 Section 4.3.5 and is the single most frequent configuration fault in the kit.
- 3.3.3** The camera is allocated the dedicated CSI port rather than the GPIO stack because image data moves at rates the general-purpose pins cannot serve. The camera requires a high-speed lane of its own. This is the reason the camera is the one component in the kit that connects by cable rather than stacking.
- 3.3.4** The stacking order of the three GPIO-mated boards follows the logic of the system: power is the foundation, the computer that consumes that power mates directly above it, and the sensors that report to that computer mate above the computer. Understanding this ordering assists fault isolation during assembly.

Table 3-2. Flight Segment Interface Definition

Device	Bus	Address / Port	Function
Waveshare UPS HAT E	I2C	0x2D	Battery voltage, current, state of charge
BME280	I2C	0x76 → 0x77	Temperature, pressure, humidity, altitude
LTR390	I2C	default	UV and ambient light
TSL2591	I2C	default	Lux, infrared, visible light
ICM20948 (IMU)	I2C	0x68	Acceleration and angular rate
SX126x LoRa	UART	/dev/serial0	Radio downlink
Pi Camera v2	CSI	ribbon cable	Imaging via rpicalm-still

CAUTION The addresses of Table 3-2 are the defaults for the reference breakout boards and are not guaranteed. Some boards ship at an alternate address; some permit the address to be changed by solder jumper. Where a scan reports a differing address, refer to MOSS-DOC-001 Section 7.2.4.

4.0 COMPONENT IDENTIFICATION

4.0.1 This section identifies each component of the flight segment by illustration and states its interface parameters. The operator shall verify receipt of every item of Table 3-1 against this section prior to commencing the assembly of Section 5.0.

4.1 Raspberry Pi 3 Model B+

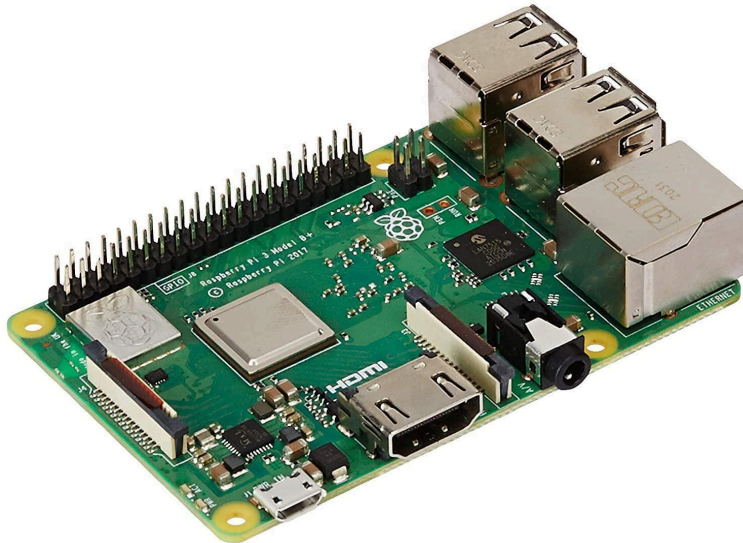


Figure 4-1. Raspberry Pi 3 Model B+

Table 4-1. Raspberry Pi 3 Model B+ Interface Parameters

Parameter	Value
Role	Flight computer
Interfaces presented	40-pin GPIO header; CSI camera port; UART; microSD
Power input	Supplied by the UPS HAT through the GPIO header
Reference OS	Raspberry Pi OS based on Debian Trixie
Flight rated	No

4.2 Waveshare UPS HAT E



Figure 4-2. Waveshare UPS HAT E

Table 4-2. Waveshare UPS HAT E Interface Parameters

Parameter	Value
Role	Power management and battery housing
Bus / address	I2C, 0x2D
Reported telemetry	Battery voltage, current, state of charge
Cells	Li-ion, installed by the operator; polarity critical
Controls	Onboard power switch; shall remain OFF until 5.6
Flight rated	No

4.3 Waveshare Sense HAT (B)

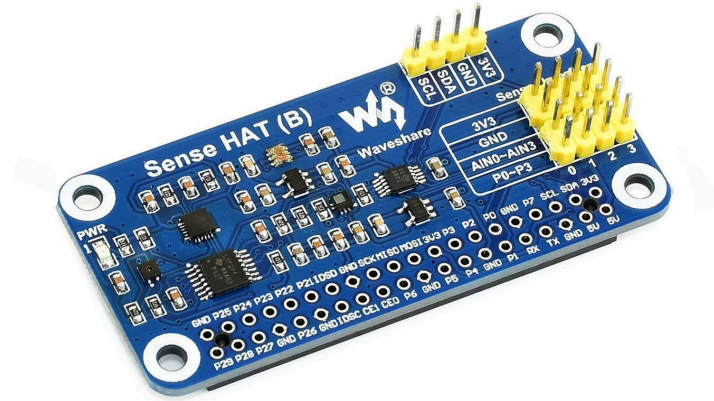


Figure 4-3. Waveshare Sense HAT (B)

Table 4-3. Waveshare Sense HAT (B) Interface Parameters

Parameter	Value
Role	Consolidated sensor suite (baseline configuration)
Bus	I2C, via the 40-pin GPIO header
Sensors	Temperature, humidity, barometric pressure, IMU (accelerometer, gyroscope, magnetometer)
Processing	None. All data is read by the flight computer.
Flight rated	No

4.4 Discrete Sensor Suite (ATLAS-1 Reference Configuration)

4.4.1 Where the discrete configuration is employed in place of the Sense HAT, the four breakout sensors of Table 4-4 shall be wired to the common I2C bus. Each device initializes independently in software; the failure of one device does not suppress acquisition from the remainder.

Table 4-4. Discrete Sensor Interface Parameters

Device	Address	Measurements
BME280	0x76 → 0x77	Temperature, barometric pressure, relative humidity; altitude derived from pressure
LTR390	default	UV index and ambient light
TSL2591	default	Lux, infrared, and visible light
ICM20948	0x68	Three-axis acceleration and angular rate; magnetic field

NOTE The LTR390 and TSL2591 reside at fixed default addresses and are located automatically by their drivers. No operator address configuration is required for these two devices.

4.5 SX126x LoRa Transceiver

Table 4-5. SX126x LoRa Transceiver Interface Parameters

Parameter	Value
Role	One-way telemetry downlink
Bus / port	UART, /dev/serial0 (exclusive use required)
Reference frequency	915 MHz
Reference transmit power	22 dBm
Packet	50-byte binary payload, schema v2, transmitted twice per record
Flight rated	No

CRITICAL The radio parameters shall be identical on the flight and ground segments. Radios configured with differing parameters will not communicate, and no error is reported. Refer to MOSS-DOC-001 Section 7.4.

4.6 Raspberry Pi Camera Module v2

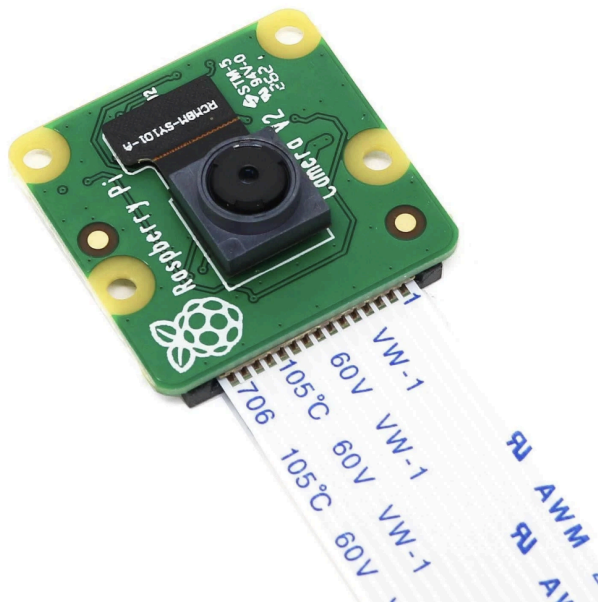


Figure 4-4. Raspberry Pi Camera Module v2

Table 4-6. Camera Module v2 Interface Parameters

Parameter	Value
Role	Imaging payload
Bus / port	CSI, by ribbon cable; does not stack
Capture tool	rpicam-still (Debian Trixie); libcamera-still on earlier releases
Reference interval	25 seconds between exposures
Flight rated	Yes

4.7 Structural Frame (1U CubeSat Mockup)

4.7.1 The stack of Section 3.1 may optionally be housed within a 1U CubeSat-format mockup frame. The frame presents the flight segment in a form factor visually representative of an orbital CubeSat for outreach, demonstration, and student design training. The frame is not a load-bearing member of the HAB payload and is not required for balloon flight; the stack and its standoffs, per Section 5.0, remain the sole structural path of the flight configuration.

4.7.2 The reference mockup assembly comprises five printed parts: a top frame, a bottom frame, two side panels, and a mid support that braces the enclosure around the stack. The parts are modeled as a single CAD assembly and are dimensioned to enclose the stack of Section 3.1 within a nominal 1U envelope.

Table 4-7. Frame Mockup Assembly Parts

Part	Function
Top Frame	Closes the upper face of the enclosure and locates the top-face standoff points
Bottom Frame	Closes the lower face and provides the base on which the stack of Section 3.1 rests
Side Panel 1	Encloses one lateral face; mates to the top and bottom frames
Side Panel 2	Encloses the opposing lateral face; mates to the top and bottom frames
Mid Support	Internal brace between the top and bottom frames; prevents wall flex and supports the stack laterally

4.7.3 The reference parts are additively manufactured. Section 7.0 states the design guidance and print procedure by which a student team may fabricate its own frame.

5.0 MECHANICAL ASSEMBLY PROCEDURE

5.0.1 This section prescribes the mechanical assembly of the flight segment. The procedure shall be performed in the sequence stated. The sequence is not arbitrary: it follows the interface allocation rationale of Section 3.3, and departure from it will require partial disassembly.

5.0.2 This section addresses mechanical assembly only. No software is executed and no interface is enabled at this stage. Following completion of this section, the operator shall proceed to MOSS-DOC-001 Section 4.0 for operating system preparation and interface enablement.

5.1 Prerequisites

5.1.1 The operator shall establish the conditions of Table 5-1 before commencing assembly.

Table 5-1. Assembly Prerequisites

Condition	Requirement
Work surface	Flat and ESD-safe. An anti-static mat or a wooden table is acceptable. Carpet shall not be used.
UPS HAT power switch	Shall be OFF.
Batteries	Shall not be installed until directed by 5.2.
Component layout	All components of Table 3-1 shall be laid out and visually identified against Section 4.0 before any connection is made.
Tools	Small Phillips driver for the standoff screws.

CAUTION Electrostatic discharge damages semiconductors invisibly. A board damaged by ESD may pass a bus scan and fail in flight. The operator shall ground themselves against a metal object before handling any board, and shall handle boards by the edges.

5.2 Step 1 — Build the Power Base

5.2.1 Every board in the stack requires stable power, so the UPS HAT (Figure 4-2) constitutes the foundation upon which the stack is built.

5.2.2 Install the Li-ion cells into the holders on the UPS HAT, observing polarity (+/–) as marked on the board.

5.2.3 The power switch shall remain OFF.

5.2.4 Attach the mounting standoffs that will carry the Raspberry Pi above the UPS HAT.

CRITICAL A cell installed in reverse polarity will damage the UPS HAT. Polarity shall be verified against the board marking before the cell is seated, and shall be re-verified before the switch is placed to ON in 5.6.



Figure 5-1. UPS HAT with cells installed and standoffs attached

5.3 Step 2 — Seat the Flight Computer

5.3.1 Lower the Raspberry Pi (Figure 4-1) onto the UPS HAT such that the Pi GPIO header aligns with the UPS HAT connector pins.

5.3.2 Align every pin before applying any pressure. Press down evenly on all four corners.

5.3.3 Secure the Pi to the standoffs with the screws provided.

5.3.4 Rationale. The flight computer mates directly to the power base because it is the board with the largest power demand and therefore warrants the shortest, most direct connection to its source.

CAUTION Uneven pressure is the most frequent cause of a bent GPIO pin. A bent pin may short two lines of the bus, which presents in software as a sensor that does not enumerate. The operator shall not force a board that does not seat freely.

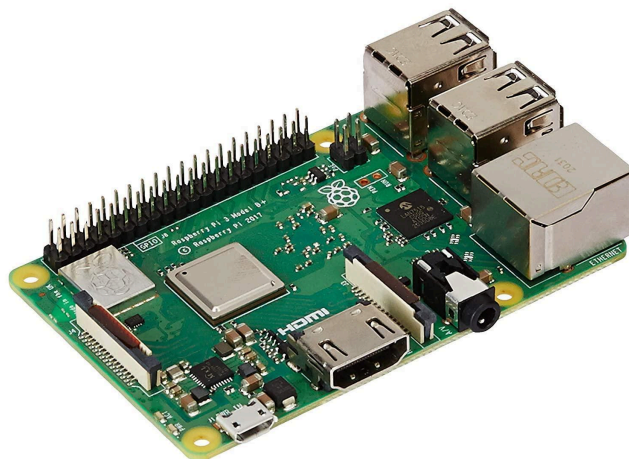


Figure 5-2. Raspberry Pi seated and secured to the UPS HAT

5.4 Step 3 — Mate the Sensor Board

- 5.4.1** Baseline configuration. Mate the Sense HAT (Figure 4-3) to the Raspberry Pi 40-pin GPIO header, in the same manner by which the Pi mated to the UPS HAT beneath it.
- 5.4.2** Align the pins carefully. The board shall sit flush and level, not tilted. Press down evenly until fully seated.
- 5.4.3** Where stacking standoffs are supplied, install them now to maintain the connection under flight vibration.
- 5.4.4** Discrete configuration. Where the discrete sensor suite of 4.4 is employed in place of the Sense HAT, wire each breakout to the common I2C bus per Table 3-2. Every device connects to the same two bus pins and is distinguished by address.
- 5.4.5** Rationale. The sensors mate directly to the GPIO header rather than connecting by cable because a HAT is designed to sit Hardware Attached on Top of those pins, affording a direct line to the flight computer for both power and data.

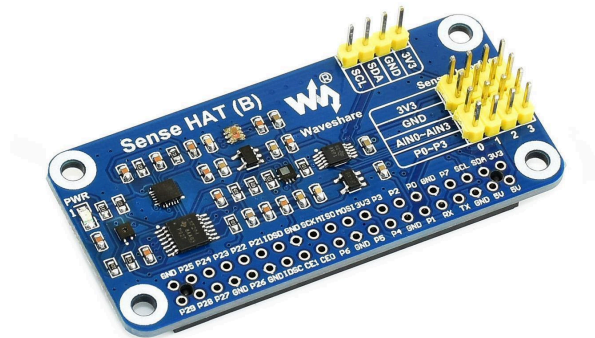


Figure 5-3. Sense HAT mated to the Raspberry Pi GPIO header

5.5 Step 4 — Connect the Camera

- 5.5.1** The camera (Figure 4-4) is the one component that does not stack. It connects by ribbon cable to the dedicated CSI port, located adjacent to the HDMI port on the Raspberry Pi.
- 5.5.2** Release the CSI port by drawing the small plastic clip upward until it lifts.
- 5.5.3** Insert the ribbon cable with the metal contacts oriented correctly. Consult the port marking or the Raspberry Pi documentation for orientation.
- 5.5.4** Press the clip down to lock the cable. The cable shall be snug and shall not withdraw under light tension.
- 5.5.5** Rationale. The camera is allocated a dedicated high-speed interface because image data moves at rates the general-purpose pins cannot serve. Refer to 3.3.3.

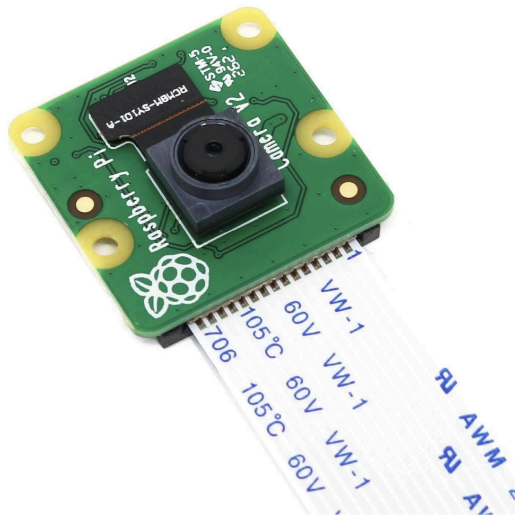


Figure 5-4. Camera ribbon cable locked into the CSI port

5.6 Step 5 — Inspection and First Power Application

5.6.1 The operator shall perform the complete inspection of Table 5-2 before power is applied. Each item eliminates one failure mode.

Table 5-2. Pre-Power Inspection

Item	Criterion
GPIO pins	No pin bent, displaced, or bridged.
Stack seating	The Pi and the sensor board are fully seated. No visible gap at any corner.
Camera ribbon	Locked in the CSI port, correctly oriented, not twisted or creased.
Cell polarity	Each cell orientation matches the board marking.
Standoffs	All screws present and firm; the stack does not flex.
Foreign objects	No loose hardware, swarf, or tooling remains within the stack.

5.6.2 Upon satisfactory inspection, place the UPS HAT power switch to ON.

5.6.3 Expected outcome: indicator LEDs illuminate on the UPS HAT and the Raspberry Pi commences boot.

CAUTION Where no board powers on, the operator shall place the switch to OFF and return to Table 5-2. Cell polarity and GPIO seating account for the majority of such indications. The operator shall not force any connection and shall not repeat the power application without first identifying a cause.

5.7 Assembly Sequence Summary

Table 5-3. Prescribed Assembly Sequence

Step	Component	Interface	Reference
1	Waveshare UPS HAT E	Base of the stack; supplies power	5.2
2	Raspberry Pi 3 Model B+	Mates to the UPS HAT GPIO connector	5.3
3	Sense HAT (B) or discrete suite	Mates to the Pi 40-pin GPIO header / I2C bus	5.4
4	Camera Module v2	Ribbon cable to the CSI port	5.5
5	Inspection and power-on	—	5.6

IMPORTANT Mechanical assembly is complete. The payload is not yet operable. The operator shall proceed to MOSS-DOC-001 Section 4.0 for operating system preparation, interface enablement, and driver installation. No sensor will enumerate and the radio will not transmit until those procedures are complete.

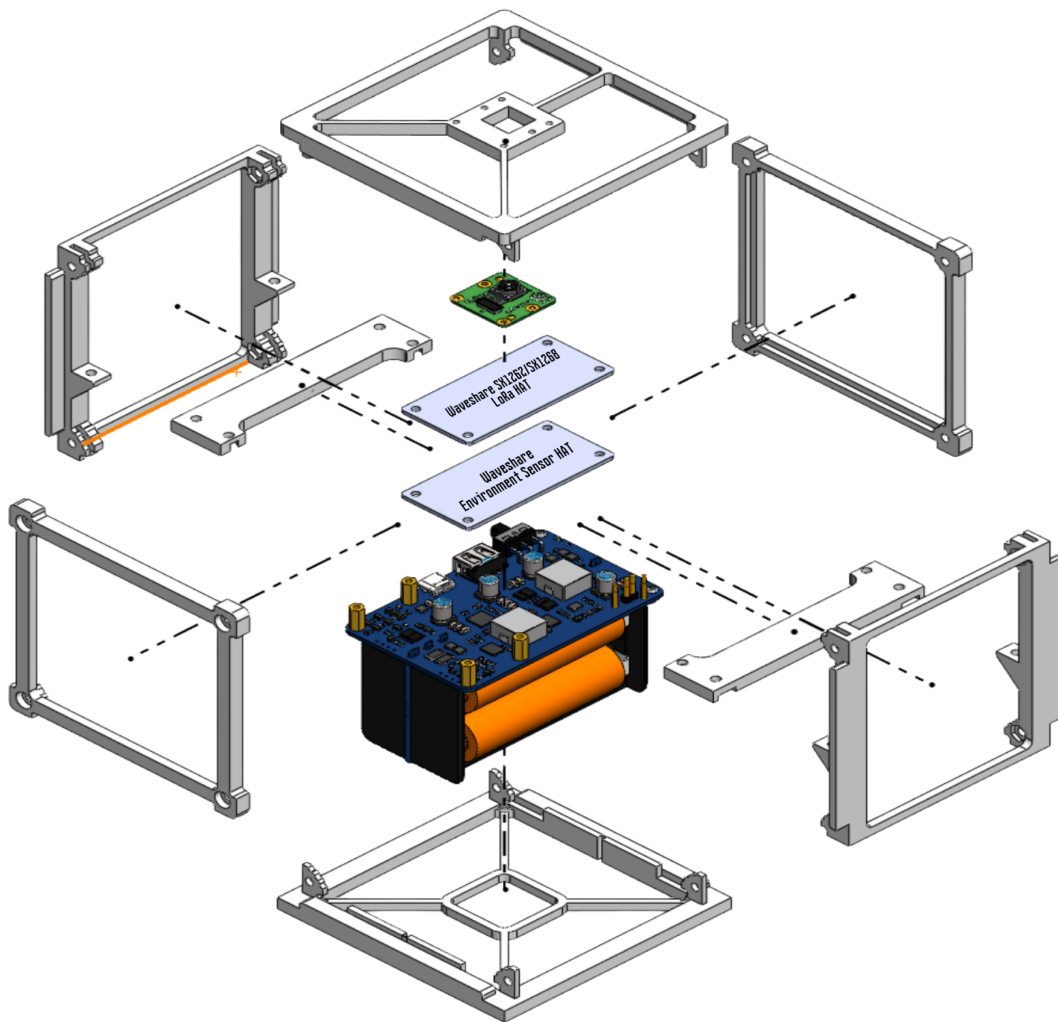


Figure 5-5. Diagram of components mentioned in table 5-3

6.0 CONTINGENCY PROCEDURES

6.0.1 This section consolidates the fault conditions arising from mechanical assembly. The indications of Table 6-1 are ordered by frequency of occurrence. Faults arising from software configuration are addressed by MOSS-DOC-001 Section 8.0.

Table 6-1. Assembly Fault Isolation

Indication	Probable Cause	Corrective Action
No board powers on	Cell polarity reversed, or GPIO not fully seated	5.6.3, Table 5-2
Sensor absent from bus scan	Bent GPIO pin, incomplete seating, or I2C not enabled	5.3.2; MOSS-DOC-001 §4.3.4
Sensor reports at an unexpected address	Board shipped at an alternate address or jumper set	4.4; MOSS-DOC-001 §7.2.4
Image capture fails	Ribbon cable reversed, not locked, or creased	5.5.3, 5.5.4
Radio never transmits, no error reported	Serial login shell enabled; OS holds the port	MOSS-DOC-001 §4.3.5
Stack flexes or separates under load	Standoffs omitted or not tightened	5.2.4, 5.4.3
Intermittent sensor dropout in flight	Stacking standoffs omitted; vibration unseats the board	5.4.3

6.1 Fault Reporting

6.1.1 The MOSS Kit is developed by students for students. Questions, defect reports, and proposed improvements shall be directed to the address below.

Item	Value
Program	UNLV ATLAS-1
Contact	TBD
Document	MOSS-DOC-002, Rev. A

7.0 3D-PRINTED 1U CUBESAT FRAME

7.0.1 This section states the design and fabrication guidance for the 1U CubeSat-format mockup frame identified in 4.7. It is provided so a student team may reproduce or modify the frame using a fused-deposition-modeling (FDM) desktop 3D printer. As stated in 4.7.1, the frame is an outreach and demonstration enclosure; it is not qualified as a load-bearing flight structure and shall not be represented as flight rated.

7.1 1U Form Factor Reference

7.1.1 The reference standard for a flight-rated 1U CubeSat exterior envelope is 100 mm by 100 mm by 113.5 mm, per the CubeSat Design Specification maintained by Cal Poly SLO. The MOSS Kit frame follows this envelope as a visual and dimensional reference only; the standard's rail, rail-pin, and deployment-switch requirements govern deployment from a P-POD launch canister and do not apply to a HAB payload, which is neither rail-mounted nor canister-deployed.

7.1.2 The stack of Section 3.1, being built from a Raspberry Pi and two stacking HATs on a 40-pin header, is taller and narrower than the internal volume a rail-mounted 1U structure normally allocates to payload. The frame parts of Table 4-7 therefore depart from the rail-and-rail-pin geometry of the standard and instead form a simple four-wall shell sized to the stack, with mounting bosses positioned to match the stack's standoff pattern.

7.2 Material Selection

7.2.1 PETG is the recommended filament for all five parts. It prints with less warping than ABS on an unheated or lightly heated bed, tolerates the temperature swing of a HAB ascent and descent better than PLA, and does not require a heated enclosure.

7.2.2 PLA may be used for a ground-display mockup that will not fly, at the cost of reduced heat resistance. PLA shall not be used for a frame intended for balloon flight, as the payload bay of an ascending HAB can exceed the temperature at which PLA begins to soften.

7.2.3 ASA or ABS may be substituted where greater UV and thermal resistance is desired, at the cost of a heated enclosure being recommended to control warping, particularly for the Top Frame and Bottom Frame V2 parts, which present the largest flat surface area.

7.3 Print Settings and Orientation

Table 7-1. Reference Print Settings

Parameter	Reference Value
Layer height	0.2 mm
Wall count	3 perimeters minimum for the Side Panels and FrameMid Support; 4 for the Top Frame and Bottom Frame V2, which carry the mounting bosses
Infill	15–20 percent, gyroid or grid pattern

Parameter	Reference Value
Print orientation	Top Frame and Bottom Frame V2 flat face down on the bed; Side Panels and FrameMid Support standing on their narrow edge to align layer lines with the primary load direction
Supports	Not normally required; add supports under any boss or standoff feature that overhangs more than 45 degrees

CAUTION Layer lines are weakest in the build (Z) direction. Orienting the Side Panels and FrameMid Support to stand on edge, rather than printing flat, places the layer lines across the panel rather than along it, which is the more frequent direction of load during handling and ground transport.

7.4 Post-Processing and Assembly

- 7.4.1** Remove all support material and clean each mounting boss with a hand reamer or an appropriately sized drill bit turned by hand, so that the standoff screws of Section 5.2 pass freely without cutting new threads in the printed plastic.
- 7.4.2** Dry-fit the Bottom Frame V2, Side Panel 1, Side Panel 2 V2, and FrameMid Support before fastening. All four parts shall seat flush without forcing; a joint that does not close under light hand pressure indicates a dimensional error that shall be corrected before final assembly.
- 7.4.3** Complete the stack assembly of Section 5.0 first, then lower the completed stack into the Bottom Frame V2, install the Side Panels and FrameMid Support around it, and close the enclosure with the Top Frame. Assembling the stack before the frame allows every GPIO connection to be inspected per Table 5-2 without the enclosure obstructing view of the header.
- 7.4.4** Where the camera of 5.5 is installed, position its opening in the printed shell before final closure, and route the ribbon cable clear of any wall joint so that it is not pinched when the Top Frame is fastened.

NOTE Because the frame is not a load-bearing member per 4.7.1, a fastening error in the enclosure does not compromise the mechanical integrity of the payload. The stack's own standoffs, inspected under Table 5-2, remain the governing structural check regardless of whether a frame is fitted.

NOTICE This document is a working release. Content is complete for the procedures described; sections not yet released are identified as TBD. Comments on this document shall reference the section number.